INCH-POUND

MS27391D
24 September 2003
SUPERSEDING
MS27391C
23 August 86

DETAIL SPECIFICATION SHEET

NIPPLE, FLARELESS, TUBE TO HOSE - SWIVEL NUT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-27272

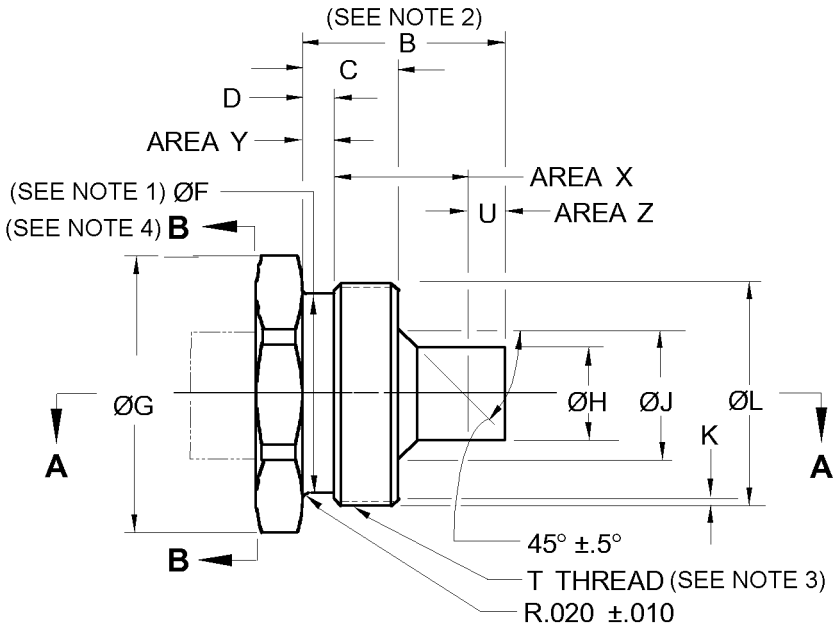

FIGURE 1. Nipple illustration.

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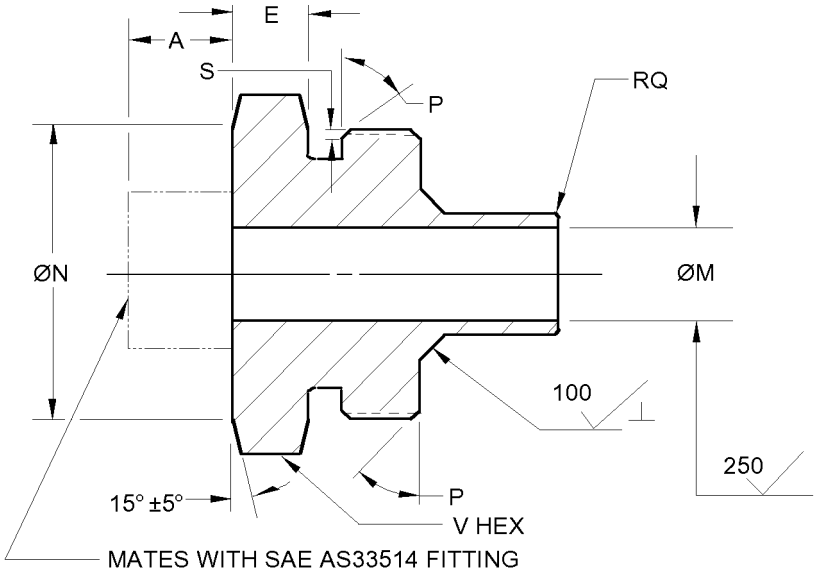

SECTION A-A

NOTES:

1. Thread gage must enter thread relief (sizes -10 through -24 only).
2. All diameters within B dimension shall be concentric within .005 full indicator movement.
3. Threads shall be in accordance with SAE AS8879 except for -8 and -24 sizes, which shall be in accordance with ANSI ASME B1.1. Threads shall be rolled on corrosion-resistant steel only.
4. Any design to the left of plane B-B is acceptable provided the dimension A and the requirements of this specification sheet and the procurement specification are met.

FIGURE 1. Nipple illustration - Continued.

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REQUIREMENTS

Intended use. This part is a component of MS27386. This is a design standard for manufacturing purposes. The item is only procured as an integral part of adapter assemblies.

Identification of product. The Part or Identifying Number (PIN) for this part shall be as specified in table I (e.g., MS27391-4C).

Dimensions and tolerances. Dimensions are in inches. Unless otherwise specified, break or radius all corners .005, +.005, -.000. All diameters must be concentric within .010 full indicator movement.

Material. PIN suffix C, corrosion-resistant steel, class 304, condition A, in accordance with SAE AMS-QQ-S-763.
PIN suffix D. Aluminum alloy, 6061-T651, in accordance with SAE AMS-QQ-A-225/8.

Finish. Corrosion-resistant steel, passivate in accordance with SAE AMS-QQ-P-35. Dry-film lubricate area X with lubricant conforming to SAE AS1701. Dry-film lubrication allowed in area Y. No dry-film lubrication allowed in area Z.
Aluminum alloy. Anodize in accordance with MIL-A-8625, type II, dye yellow.

Surface roughness. Unless otherwise specified, maximum surface roughness shall not exceed 125 µin. R_a in accordance with ASME B46.1.

Order of precedence. This specification takes precedence over the documents referenced herein. Unless otherwise specified, referenced documents shall be of the issue in effect on the date of solicitation.

Remove all burrs and slivers.

Nipple illustration. See figure 1.

TABLE I. Nipple requirements.

PIN MS27391		A ±.010	B 1/ ±.005		C ±.005		D ±.005	
Steel	Alum		Steel	Alum	Steel	Alum	Steel	Alum
-3/-4C	-	.533	.735	-	.255	-	.075	-
-4C	-	.518		-		-		-
-5C	-	.588	.730	-	.250	-	.085	-
-6C	-	.617		-		-	.050	-
-8C	-8D	.717	.870	.865	.270	.265	.085	.080
-10C	-10D	.824	.950	.957	.300	.307	.100	.107
-12C	-12D	1.028	1.035	1.018	.360	.343		.083
-16C	-16D	1.028	1.120	1.100	.390	.370	.110	.090
-20C	-20D	1.110	1.335	1.341	.400	.406	.085	.091
-24C	-24D	1.367	1.480	1.480	.500	.500	.100	.100

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TABLE I. Nipple requirements - Continued.

PIN MS27391 Steel	PIN MS27391 Alum	E ±.005 Steel	E ±.005 Alum	F 2/		G		H +.000 -.005	J ±.005	K	
-3C/4C	-	.125	-	.400	±.005	.620	±.005	.210	.295	.031	±.005
-4C	-		-								
-5C	-		-	.430		.675	+.015 -.000	.273	.360		
-6C	-	.135	-	.495		.750		.335	.425	.015	max
-8C	-8D	.150	.155	.600		.960		.431	.530	.031	±.005
-10C	-10D	.170	.163	.785	±.010	1.105		.531	.625	.035	
-12C	-12D	.190	.207	.910		1.385		.655	.760		
-16C	-16D	.220	.240	1.277		1.680		.905	1.040	.038	
-20C	-20D	.280	.274	1.589		1.979		1.156	1.275		
-24C	-24D		.280	1.828		2.340		1.406	1.550	.043	

PIN MS27391 Steel	PIN MS27391 Alum	L		M		N ±.02	P ±5º Steel	P ±5º Alum	Q		S	
-3C/-4C	-	.500	+.000 -.006	.161	+.005 -.000	.53	45º	-	.015	+.005 -.000	.031	±.005
-4C	-							-				
-5C	-	.562	+.000 -.007	.224		.59		-				
-6C	-	.625		.261		.66	30°	-			.015	max
-8C	-8D	.750		.345	+.006 -.000	.84	45°	30°	.020		.031	±.005
-10C	-10D	.875	+.000 -.008	.440		.97				±.005	-	-
-12C	-12D	1.000		.560		1.22			.030		-	-
-16C	-16D	1.375	+.000 -.009	.828		1.47					-	-
-20C	-20D	1.688		1.058		1.78					-	-
-24C	-24D	1.938	+.000 -.010	1.282	+.005 -.000	2.09			.035		-	-

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TABLE I. Nipple requirements - Continued.

PIN MS27391		T 3/		U		V ±.02
Steel	Alum	Thread	Pitch diameter	Steel only		
-3C/-4C	-	.5000 - 28 UNJEF - 3A	.4768/.4740	.25	±.05	.56
-4C	-					
-5C	-	.5625 - 24 UNJEF -3A	.5354/.5325			.62
-6C	-	.6250 - 24 UNJEF -3A	.5979/.5949			.69
-8C	-8D	.7500 - 24 UNS -3A	.7229/.7198	.32	±.12	.88
-10C	-10D	.8750 - 20 UNJEF -3A	.8425/.8392	.35	±.15	1.00
-12C	-12D	1.0000 - 20 UNJEF -3A	.9675/.9641			1.25
-16C	-16D	1.3750 - 18 UNJEF - 3A	1.3389/1.3353	.39	±.19	1.50
-20C	-20D	1.6875 - 18 UNJEF - 3A	1.6514/1.6476	.48	±.28	1.81
-24C	-24D	1.9375 - 16 UN - 3A	1.8969/1.8929	.50	±.30	2.12

1/ All diameters within B dimension shall be concentric within .005 full indicator movement.
2/ Thread gage must enter thread relief (sizes -10 through -24 only).
3/ Threads shall be in accordance with SAE AS8879 except for -8 and -24 sizes, which shall be in accordance with ANSI ASME B1.1. Threads shall be rolled on corrosion-resistant steel only.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 99
DLA - CC

Review activities:
Army - AT, MI
Navy - MC, SA, SH
Air Force - 71

Preparing activity:
DLA - CC

(Project 4730-0868-073)

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